

A closed-form variance floor for slope-capped diffusion samplers on thin manifolds

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Claim. For data in a σ -tube about a manifold, any reverse-SDE sampler whose learned normal score slope is bounded by $\bar{\kappa}$ retains normal variance at least $1/(8\bar{\kappa}^2)$ as $\sigma \rightarrow 0$, for *every* truncation time and *every* schedule. So a slope-capped model cannot track a thinning manifold, and bounded relative error forces $\bar{\kappa} \gtrsim 1/\sigma$, i.e. a slope budget growing like $\sqrt{J}$ in the Fisher information. This bounds the network’s normal Lipschitz norm, not its parameter count: a fixed architecture attains any $\bar{\kappa}$ by enlarging its weights. This note is self-contained; §3 is the entire argument (comparison principle, integrating factor, one Gamma integral).

1 Setup

Let $M \subset \mathbb{R}^d$ be a compact smooth k -manifold with reach $\tau > 0$ and $X_0 = \varphi(U) + \sigma N_\perp$, U uniform on M , $N_\perp$ standard normal in the normal space. Use the variance-preserving OU process $X_t = aX_0 + sZ$, $a = e^{-t/2}$, $s^2 = 1 - e^{-t}$. Work in one normal coordinate $x \in \mathbb{R}$, treating the fiber marginal as Gaussian and decoupled from tangential position (*flat-tube idealization*; curvature enters at $O(s^2/\tau)$). Then $X_0 \sim \mathcal{N}(0, \sigma^2)$, $X_t \sim \mathcal{N}(0, \gamma^2)$ with $\gamma^2 = a^2\sigma^2 + s^2$, and the denoising target is exactly linear:

$$\varepsilon^*(x, t) = \kappa(t)x, \quad \kappa(t) = s(t)/\gamma^2(t). \quad (1)$$

Maximising (1) over $u := s^2$ gives $u^* = \sigma^2/(1 - \sigma^2)$ and

$$\kappa_{\max} = \frac{1}{2\sigma\sqrt{1-\sigma^2}} = \frac{1}{2\sigma}(1 + O(\sigma^2)). \quad (2)$$

This is the mechanism: κ rises from 0, peaks at $\approx 1/(2\sigma)$ near $t \approx \sigma^2$, and returns to 0 as $t \rightarrow 0$. The score is steep only inside a short, finite window of noise levels, and the peak diverges as the tube thins.

Let X_t be the sampler’s *own* ensemble at noise level t and define the realised normal slope

$$\hat{\kappa}(t) := \text{Cov}(X_t, \hat{\varepsilon}(X_t, t)) / \text{Var}(X_t),$$

the L^2 regression slope of the estimator under the law the sampler actually visits. No decomposition of the network is needed. The only hypothesis used below is

$$\hat{\kappa}(t) \leq \bar{\kappa} < \infty \quad \text{for all } t, \quad (3)$$

a finite normal Lipschitz constant. This bounds the network’s normal *slope*, a weight-norm quantity, not its parameter count: a fixed architecture realises arbitrarily large $\bar{\kappa}$ by scaling its weights, so nothing below lower-bounds network size. Put $\rho := 2\sigma\bar{\kappa} = \bar{\kappa}/\kappa_{\max} + O(\sigma^2)$, the single dimensionless knob: how close the network’s ceiling gets to the peak slope the data demands.

2 Two structural facts

Lemma 1 (variance ODE). *Under the reverse Euler scheme $x \leftarrow x + (-\frac{1}{2}x + \hat{\varepsilon}/s)dt + \sqrt{|dt|}\xi$ with affine estimator, the ensemble normal variance obeys, in the small-step limit,*

$$-\frac{dV}{dt} = \left(1 - \frac{2\hat{\kappa}(t)}{s(t)}\right)V + 1. \quad (4)$$

Check: with the exact slope $\hat{\kappa} = \kappa$, $V(t) = \gamma^2(t)$ solves (4) identically. The forcing +1 is the sampler’s own injected noise.

Lemma 2 (Lipschitz sufficiency). *If $\hat{\varepsilon}(\cdot, t)$ is $\bar{\kappa}$ -Lipschitz in the normal coordinate then $\hat{\kappa}(t) \leq \bar{\kappa}$ under any law: for an i.i.d. copy X' of X , $\text{Cov}(X, g(X)) = \frac{1}{2}\mathbb{E}[(X - X')(g(X) - g(X'))] \leq \frac{\bar{\kappa}}{2}\mathbb{E}[(X - X')^2] = \bar{\kappa} \text{Var}(X)$. Lipschitz control is sufficient, not necessary.*

This is why the floor is not an artifact of linearising the score. (4) is an *identity* for an arbitrary nonlinear $\hat{\varepsilon}$: for the continuous reverse SDE $dX = f(X)d\tau + dW$ with $f(x) = \frac{1}{2}x - \hat{\varepsilon}(x, t)/s$, Itô gives $\text{Var}' = 2\text{Cov}(X, f(X)) + 1$, which is (4). The nonlinearity is not discarded, it is absorbed into the definition of $\hat{\kappa}$.

Singular rescaling. Put $t = \sigma^2 u$, $V = \sigma^2 W$, $\hat{\kappa} = (1/\sigma)m(u)$. Since $s = \sqrt{t}(1 + O(t))$, (4) becomes, as $\sigma \rightarrow 0$ at fixed ρ ,

$$-\frac{dW}{du} = 1 - \frac{2m(u)}{\sqrt{u}}W, \quad (5)$$

which is σ -free. The uncapped equation $-dW/du = 1 - 2W/(1+u)$ has general solution $W = (1+u) + A(1+u)^2$, so deviations from the slave $W = 1+u$ contract like $((1+u)/(1+u'))^2$ under downward integration: the band-entry boundary layer is suppressed quadratically. Hence $\lim_{t_0 \rightarrow 0} V(t_0)/\sigma^2 = g(\rho)(1 + O(\sigma^2/\rho^2))$, a *relative* correction.

3 The unconditional floor

Theorem 1. *Assume only (3). Then for every $t_0 > 0$ and every schedule,*

$$V(t_0) \geq \sigma^2 \frac{1}{2\rho^2}(1 + o(1)) = \frac{1}{8\bar{\kappa}^2}(1 + o(1)) \quad (\sigma \rightarrow 0), \quad (6)$$

and the floor exceeds the true width σ^2 exactly when $\rho < 1/\sqrt{2}$.

Proof. (i) *Comparison.* In (4) forcing and data are common and $\partial(-dV/dt)/\partial\hat{\kappa} = -2V/s \leq 0$: smaller slope gives larger variance. So any profile with $\hat{\kappa} \leq \bar{\kappa}$ satisfies $V \geq V_{\hat{\kappa}=\bar{\kappa}}$ pointwise. This is one-sided; no lower bound on $\hat{\kappa}$ is used, so arbitrary overshoot is permitted. (ii) *Envelope.* Take $m \equiv \rho/2$ in (5); the integrating factor $e^{-2\rho\sqrt{u}}$ gives $\frac{d}{du}[We^{-2\rho\sqrt{u}}] = -e^{-2\rho\sqrt{u}}$. (iii) *Integrate.* For data of sub-quadratic growth at $u = \infty$,

$$W(0^+) = \int_0^\infty e^{-2\rho\sqrt{u}} du = 2 \int_0^\infty xe^{-2\rho x} dx = \frac{1}{2\rho^2},$$

exactly, for every $\rho > 0$. Undoing the rescaling gives (6); $1/(2\rho^2) > 1 \iff \rho < 1/\sqrt{2}$. It applies to the full nonlinear estimator with no extension step, since (4) is exact. $\square$

Refinement (no overshoot). If also $\hat{\kappa} \leq \kappa$ (the network never over-contracts), then $V(t_0) \geq \sigma^2 g(\rho)(1 + o(1))$ with, for $q = \sqrt{1 - \rho^2}$, $x_\mp = (1 \mp q)/\rho$, $u_\mp = x_\mp^2$,

$$W_{\text{exit}} = (1 + u_+)e^{-4q} + \frac{(3-2q)-(3+2q)e^{-4q}}{2\rho^2}, \quad g(\rho) = 1 + \frac{W_{\text{exit}} - (1 + u_-)}{(1 + u_-)^2}.$$

Band edges solve $\sqrt{u}/(1+u) = \rho/2$, i.e. $\rho x^2 - 2x + \rho = 0$ ($x = \sqrt{u}$), nonempty iff $\rho < 1$; above the band $W = 1+u$ exactly, across it the same integrating factor with antiderivative $-(2\rho\sqrt{u} + 1)e^{-2\rho\sqrt{u}}/(2\rho^2)$ gives W_{exit} , and below it matching $(1+u) + A(1+u)^2$ at u_- gives $g = 1 + A$. Here $g > 1$ on $(0, 1)$ and $g = C_0/\rho^2 + O(1)$ as $\rho \rightarrow 0$ with $C_0 = (1 + 3e^{-4})/2$, so the floor is $0.1319\bar{\kappa}^{-2}$. The two constants, $1/8$ and $(1 + 3e^{-4})/8$, differ by 5.5%.

4 Consequence, assumptions, and how to refute it

Hold $\bar{\kappa}$ fixed and let $\sigma \rightarrow 0$, so $J \asymp (d - k)/\sigma^2 \rightarrow \infty$. Then $V/\sigma^2 \geq (1 + o(1))/(8\bar{\kappa}^2\sigma^2) \rightarrow \infty$: a sampler held at a fixed slope ceiling collapses without bound relative to the data. Inverting, bounded relative error needs $\bar{\kappa} \gtrsim 1/\sigma$, i.e. a slope budget $\sim \sqrt{J}$. Since the bound holds for every t_0 and every schedule, no annealing or truncation schedule evades it. Whether a given architecture supplies that slope budget is a separate question, not settled here.

Assumed (where a sceptic should push): (1) the flat-tube idealization, one normal coordinate with Gaussian fiber decoupled from tangential position. The *variance-flow* half of this is no longer assumed. Writing h for the signed distance to the manifold ($|\nabla h| = 1$), Itô gives (4) plus exactly one extra term, $\text{Cov}(h, \Delta h)$, which vanishes

identically for a flat normal coordinate and equals $(D-1)(1-\mathbb{E}[r]\mathbb{E}[1/r]) \leq 0$ on a sphere. If Δh is L -Lipschitz in h then $|\text{Cov}(h, \Delta h)| \leq LV$ and Theorem 1 holds with the coefficient shifted by $-L$; since $2\bar{\kappa}/s$ diverges as $s \rightarrow 0$ while L is constant, this costs 2–3% of the floor on our geometries and leaves the $\sigma \rightarrow 0$ law unchanged. It is verified against a closed form (Brownian motion from the origin, where the difference reduces to $z\Gamma(z) = \Gamma(z+1)$) and on trained networks, most sharply through the companion mean equation, which on S^2 holds to 0.7% at every noise level tested while the flat form is wrong by factors. What remains assumed is the Gaussian-fiber half, where curvature still enters at $O(s^2/\tau)$ and is not carried in the constants; (2) $\hat{\kappa}$ is the regression slope under the sampler’s own ensemble, so a probe that measures it under p_t instead estimates a different quantity whenever $\hat{\varepsilon}$ is nonlinear; (3) the $o(1)$ in (6) is the $\sigma \rightarrow 0$ limit, relative correction $O(\sigma^2/\rho^2)$, so the additive gap to g is $O(\sigma^2/\rho^4)$.

Checks. At <https://github.com/Naman84789/model-collapse-manifolds>, `falsify_floor.py` attacks the result from (4) (RK45 vs. both closed forms; limits $\rho \rightarrow 1^-$, $\rho \rightarrow 0$, $\bar{\kappa} \rightarrow \infty$; 200 random slope profiles including overshooting ones; boundary-layer contraction; symbolic re-derivation). `e2e_floor_check.py` attacks it *before* (4): it recovers (2) from the density, then simulates the reverse SDE directly with a slope-capped score and measures the output variance without using (4) at all, recovering $1/(2\rho^2)$ to 1.7% at the equality case.

To refute: exhibit a profile with $\hat{\kappa} \leq \bar{\kappa}$ whose measured $V(t_0)$ falls below $\sigma^2/(2\rho^2)$.